Hereditary Cardiovascular Gene Panels

Updated 22 October 2026 | Expanded version for clinical panel review

102 unique genes across five disease groups. Red + identifies each of the 16 newly added genes. All original genes are retained.

No.	Cardiomyopathy (43)	Arrhythmia (28)	Aortopathy (23)	PAH (12)	Lipid (8)
1	ACTC1	CACNA1C	ACTA2	ACVRL1 +	ABCG5
2	ACTN2 +	CACNB2	COL3A1	ATP13A3 +	ABCG8
3	ALPK3 +	CALM1	COL5A1	BMPR2	APOB
4	BAG3	CALM2	COL5A2	CAV1	APOE
5	CAV3	CALM3	EFEMP2	EIF2AK4 +	LDLR
6	CRYAB	CASQ2	ELN	ENG +	LDLRAP1
7	CSRP3	CAV3	FBN1	GDF2	LIPA
8	DES	DES	FBN2	KCNK3 +	PCSK9
9	DMD +	DSC2	FLNA	KDR +	
10	DSC2	DSG2	LOX	SMAD9	
11	DSG2	DSP	MYH11	SOX17 +	
12	DSP	GPD1L	MYLK	TBX4 +	
13	FHL1	HCN4	NOTCH1		
14	FHOD3 +	JUP	PRKG1		
15	FLNC	KCNE1	SKI		
16	GLA +	KCNE2	SLC2A10		
17	JPH2	KCNH2	SMAD2		
18	JUP	KCNJ2	SMAD3		
19	LAMP2	KCNQ1	SMAD4		
20	LMNA	LMNA	TGFB2		
21	MYBPC3	PKP2	TGFB3 +		
22	MYH7	PLN	TGFBR1		
23	MYL2	RYR2	TGFBR2		
24	MYL3	SCN5A			
25	NEXN +	TECRL +			
26	PKP2	TMEM43			
27	PLN	TRDN			
28	PRKAG2	TTN			
29	PTPN11				
30	RAF1				
31	RBM20				
32	RIT1				
33	RYR2				
34	SCN5A				
35	SGCD				
36	TCAP				
37	TNNC1				
38	TNNI3				
39	TNNT2				
40	TPM1				
41	TTN				
42	TTR				
43	VCL				
Total	**43**	**28**	**23**	**12**	**8**

Group totals include shared genes and therefore exceed the unique-gene count. Baseline: 86 unique genes; additions: 16; updated total: 102.

Source baseline: Recommended Hereditary Cardiovascular Gene Panels, saved in Readwise on 22 October 2026.

Additions & implementation notes

New genes are highlighted in red on the panel table. Priorities below distinguish core additions from phenotype-dependent extensions.

Cardiomyopathy | 6 additions

GLA: Fabry disease, an actionable LVH/HCM mimic; disease-specific therapy and family evaluation may follow a confirmed diagnosis. [1]
FHOD3, ALPK3, ACTN2: updated HCM/intrinsic cardiomyopathy coverage. FHOD3 and ACTN2 have definitive evidence; ALPK3 has strong dominant and definitive recessive evidence. [2]
DMD: dystrophin-related cardiomyopathy, including cardiac-predominant disease and affected female carriers. Require validated deletion/duplication analysis. [3]
NEXN: second-priority DCM extension; individually uncommon and dependent on phenotype and variant interpretation. [4]

Arrhythmia | 1 addition

TECRL: established recessive CPVT; particularly relevant to exercise/emotion-triggered ventricular arrhythmias, sometimes with QT prolongation. Rare overall. [5]

Aortopathy | 1 addition

TGFB3: second-priority extension for TGFB3-related heritable thoracic aortic disease/Loeys-Dietz spectrum. Evidence and outcome data are less extensive than for the core HTAD genes. [6]

PAH | 8 additions

ACVRL1, ATP13A3, EIF2AK4, ENG, KCNK3, KDR, SOX17, TBX4: together with the original four genes, these complete the 12 definitive-evidence genes in the 2023 ClinGen PAH assessment. EIF2AK4 also addresses heritable pulmonary veno-occlusive disease/capillary haemangiomatosis; ACVRL1 and ENG cover HHT-associated disease. [7]

Lipid | No additions

The original eight-gene lipid panel is retained in this update.

Laboratory & interpretation requirements

This is a proposed expanded panel, not confirmation of laboratory assay coverage. If WES is used, confirm that all listed genes are included in the analysis and that coverage and copy-number detection are validated. Retaining an original gene does not imply equal evidence across all phenotypes. Use phenotype-specific gene-disease validity and inheritance; a VUS must not drive treatment or predictive cascade testing. [3,8]

Selected references

[1] Fabry Disease - GeneReviews
[2] ClinGen HCM gene reappraisal. JACC, 2025
[3] EMQN best practice guidelines for dystrophinopathy testing, 2020
[4] NEXN in cardiomyopathies and sudden cardiac death, 2023
[5] ClinGen CPVT and short QT gene validity assessment, 2021
[6] TGFB3-related HTAD: Montalcino Aortic Consortium
[7] ClinGen PAH gene validity assessment, 2023
[8] EHRA/HRS/APHRS/LAHRS cardiac genetic testing consensus, 2022